

1 Beam parameters and first ChDR simulation case

Let us fix the beam parameters as follows, assuming 60 MeV means kinetic energy and 3 mm means the longitudinal rms size, denoted σ_x because the beam travels along the x axis in the figure:

$$E_{\text{kin}} = 60 \text{ MeV}, \quad Q = 5 \text{ nC}, \quad \sigma_x = 3 \text{ mm}.$$

The corresponding total energy, relativistic factor and speed are

$$E_{\text{tot}} = 60.511 \text{ MeV}, \quad \gamma = 1 + \frac{60}{0.510999} = 118.417, \quad \beta = \sqrt{1 - \gamma^{-2}} = 0.99996434.$$

The bunch duration is $\sigma_t = \sigma_x/c = 10.0 \text{ ps}$. A Gaussian bunch therefore has an rms-to-FWHM duration of about 23.6 ps, and its peak current is

$$I_{\text{peak}} = \frac{Q}{\sqrt{2\pi}\sigma_t} \simeq 199 \text{ A}.$$

The charge corresponds to

$$N_e = \frac{Q}{e} = 3.12075 \times 10^{10} \text{ electrons},$$

and the bunch kinetic energy is approximately 0.30 J.

The current FEL job-file parser does not interpret **charge** and **mass** directly as SI coulombs and kilograms. It converts the numerical entries from electron-charge and electron-mass units into the internal normalized units used by the solver. Therefore, for a total charge of 5 nC, the corresponding entries are

```
"charge": 3.1207545e10,
"mass": 3.1207545e10,
"gamma": 118.417071,
"sigma-position": [sigma_x_um, sigma_y_um, 3000.0]
```

The value 3 mm is 3000 μm . For the laboratory-frame ChDR model, the computational domain along the beam should cover at least 6–10 σ_x , or roughly 18 mm to 30 mm, before adding vacuum margins and the dielectric radiator.

For a Gaussian bunch the longitudinal form factor is

$$|F(\omega)|^2 = \exp\left[-\left(\frac{2\pi\sigma_x}{\lambda}\right)^2\right].$$

Thus the coherent spectrum is strongest at wavelengths of order $2\pi\sigma_x \simeq 18.8 \text{ mm}$, corresponding to about 15.9 GHz. Coherent visible emission is strongly suppressed for this bunch length; a visible camera signal would require an incoherent single-electron treatment or optical-scale microbunching. For coherent emission at fixed bunch shape, the radiated power scales approximately as Q^2 .

The figure below is a schematic laboratory-frame setup. The dielectric value $\epsilon_r = 2.13$ is illustrative, corresponding to $n = \sqrt{2.13}$ and $\theta_{\text{Ch}} \simeq 46.7^\circ$ at this beam energy. The impact parameter a , transverse beam sizes and radiator dimensions remain geometry parameters for the simulation.

2 Geometry choice and proof-of-concept scope

There is no fundamental obstacle to a proof-of-concept simulation. For $\epsilon_r = 2.13$, the value near 47° is approximately the internal Cherenkov emission angle,

$$\cos \theta_{\text{Ch}} = \frac{1}{\beta n}, \quad n = \sqrt{\epsilon_r}.$$

The dielectric surface itself does not have to be tilted by this angle. A flat dielectric slab parallel to the beam is sufficient to test radiation generation, dependence on the impact parameter a , spectrum, polarization and the internal Cherenkov angle. It does not reproduce the final out-coupling through the prism or the exact camera signal.

2.1 Limitations to be tackled by the MSc Thesis of D Rosser

The present FEL mini-app has several limitations for this problem.

1. Its field solver assumes vacuum and does not yet support a spatially varying $\epsilon_r(\mathbf{r})$, so a dielectric interface has to be added. An oblique interface on a Cartesian grid also produces staircasing and spurious scattering; an axis-aligned slab is therefore the cleaner first test. A tilted prism can later be represented with a material mask and subcell averaging or a conformal interface treatment.
2. The FEL implementation uses a Lorentz-boosted frame and treats z as the longitudinal direction. A stationary radiator requires a laboratory-frame calculation. The z axis in this figure is consequently a drawing convention unless the code is also rotated or refactored.
3. Second-order Mur boundaries may reflect oblique waves and dielectric-interface radiation; they are acceptable for an initial test, while PML boundaries are preferable for quantitative power <https://arxiv.org/abs/1105.5625>.

2.2 Experimental Considerations

1. With $\sigma_z = 3$ mm, coherent radiation is concentrated mainly around 16 GHz, so centimetre-scale domains are appropriate.
2. A visible-camera prediction would require an incoherent single-electron treatment or optical-scale bunching.

Parameter space for the first POC is therefore a straight 60 MeV, 5 nC beam passing parallel to a flat $\epsilon_r = 2.13$ slab, with the gap a varied. We can validate the spectrum, polarization and internal 46.7° radiation angle before implementing the tilted prism and optical extraction.

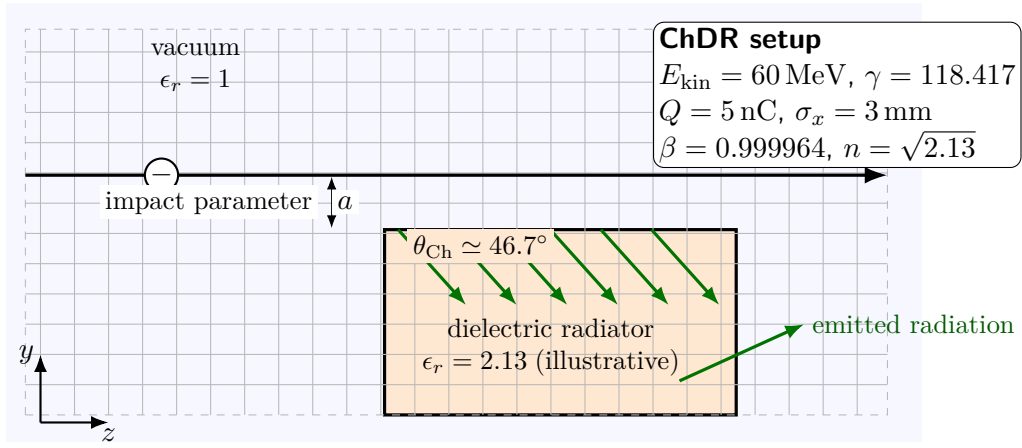


Figure 1: Schematic x - z view of the Cherenkov-diffraction-radiation setup. The continuous grid represents the computational domain in both vacuum and dielectric.